

A Tkinter GUI application that fetches and displays real-time weather, an hourly forecast, and a 5-day forecast using the Open Weather Map API.

SETUP:

1. Register for a free API key at <https://openweathermap.org/api>

(New keys can take ~10-15 minutes to activate.)

2. Either:

- a) Set it as an environment variable: `export OPENWEATHER_API_KEY="your_key_here"`
- b) Or paste it directly into the API_KEY constant below.

3. Install dependencies:

```
pip install requests pillow
```

4. Run:

```
python advanced_weather_app.py
```

Notes on the free tier:

- The "5 day / 3 hour" forecast endpoint is used for both the hourly and daily panels (the free tier does not offer true per-hour or per-day endpoints). The hourly panel shows the next 6 available 3-hour slots (i.e. the next ~18 hours in 3-hour steps) labelled by their timestamp, and the daily panel picks one representative (mid-day) reading per day for the next 5 days.
- Auto-location uses ip info. io's free endpoint (no key required for light/basic use) to guess the user's city from their public IP. This is approximate and can be wrong for users on VPNs or mobile networks.